Many Daughters: Pre-Analysis Plan

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1 General Interpretation of Hypotheses and General Econometric Strategy

1.1 General Interpretation of Hypotheses

All hypotheses share a common beginning: "Having daughters leads to [..]". We are torn between two possible interpretations:

1. The interpretation of "having daughters" as representing individuals or households with at least one daughter, as it is the common interpretation in English.

2. Restricting the treatment group to individuals or households that have at least two daughters, since a) the project title implies "Many Daughters", b) the plural "daughters" could be interpreted meaning actually "more than one".

We decided to pick interpretation (1), meaning our treatment group will consist of individuals or households that have at least one daughter. In the end, we think this is the scientifically more interesting and relevant question.

We understand the term "having" as referring to being a parent. We therefore restrict the sample to private households, based on the variable pop. While we believe that smaller public households could in principle be affected by similar effects of having daughters as private households, the number of girls or daughters in the households is not as strictly exogenous as in the case of (biological) parents, as discussed in the next section.

1.2 General Econometric Strategy

In this section, we provide our general econometric strategy, as well as information on variables used in all regressions.

In Germany, it is illegal to have gender-specific abortions. Therefore, as long as parents have K_i kids, we treat having daughters as being uncontrollable and therefore exogenous to the parents.[1] Consequently, coefficients that result from estimating the following regression equation:

$$y_i = \beta_0 + \beta_1 D_i + \beta_K K_i + \epsilon_i \tag{1}$$

will not suffer from endogeneity issues such as reverse causality or omitted variable bias, where y_i is the dependent variable of choice and i either means individual or household.

Since the exact number of daughters is not part of the hypotheses, we encode D_i as a binary variable, taking on the value of one if an individual or household has at least one daughter and

[1] A possible threat to this assumption are adoptions. Since the SOEP does not provide information on adoptions, we have no way of correction. However, adoptions are rare: in 2024, only 3662 kids were adopted in Germany, compared to the number of births of 677,117. 74% of adopted kids were former stepchildren. (Destatis, 2025). We treat adoptions as measurement error.

zero otherwise:

$$D_i = \begin{cases} 1 & \text{if individual or household } i \text{ has at least one daughter} \\ 0 & \text{if individual or household } i \text{ has no daughters} \end{cases}$$

In the SOEP data, the gender of kids is given by the variables kidsex01 to kidsex19. For the total number of kids K_i, we use the variable sumkids.

Since the model given in equation (1) is the most parsimonious that will yield unbiased results, we refrain from including additional controls or fixed effects. The interpretation of coefficients will then be that of a pooled OLS estimator, that exploits both the cross-sectional and within variation in the data. We will keep this basic structure for testing hypotheses H1, H2 and H3. For H4, we include interaction effects with respect to gender.

2 H1: Gender Equality as Measured by Attitudes

The first hypothesis to test is:

Having daughters leads to stronger support for gender equality (as measured by attitudes).

2.1 Variables

As far as we are aware, the SOEP data does not provide a large amount of survey questions regarding the attitudes towards gender equality. The variables we found to be in close alignment with our understanding of attitudes towards gender equality[2] additionally have the problem of being mostly restricted to one or two survey waves. We found that a combination of the variables plh0308_v1 and plh0308_v2 provides the best compromise between being close to our interpretation of gender equality and sample size. Both variables are composed of the answers to the same question: *"Am besten ist es, wenn der Mann und die Frau beide gleich viel erwerbstätig sind und sich beide in gleichem Maße um Haushalt und Familie kümmern."*. However, answers differ between the variables, i.e. plh0308_v1 has answer-levels 1-4, where 4 means complete disagreement, and plh0308_v2 has 7 answer-levels, where 7 means complete agreement. To make answers compatible, we construct our final dependent variable as follows:

1. Flag all negative values of plh0308_v1 and plh0308_v2 as missing.

2. Reverse ordering of plh0308_v1:

$$\text{plh0308_v1} = 5 - \text{plh0308_v1} \quad \text{if plh0308_v1}\neq\text{missing}$$

[2] A question that comes close to our understanding of gender equality would have been "Befürworten Sie die vollständige de-facto Gleichberechtigung von allen Personen, unabhängig vom Geschlecht?"

3. Adjust answer-levels of plh0308_v2 to be in line with plh0308_v1

$$\text{plh0308_v2} = \begin{cases} 1 & \text{if plh0308_v2} \in \{1, 2\} \\ 2 & \text{if plh0308_v2} \in \{3, 4\} \\ 3 & \text{if plh0308_v2} \in \{5, 6\} \\ 4 & \text{if plh0308_v2} = 7 \end{cases}$$

4. Merge adjusted variables plh0308_v1 and plh0308_v2 together to form our measure of gender equality (attitudes), GEA_i:

$$\text{GEA}_i = \begin{cases} \text{plh0308_v1} & \text{if plh0308_v1}\neq\text{missing} \\ \text{plh0308_v2} & \text{if plh0308_v2}\neq\text{missing} \end{cases}$$

2.2 Exact Analysis and Interpretation of Results

We estimate the following regression equation:

$$\text{GEA}_i = \beta_0 + \beta_1 D_i + \beta_K K_i + \epsilon_i$$

where is GEA_i is our measure of attitudes towards gender equality as discussed in the previous section, D_i is a binary variable takes on the value of 1 if an individual has at least two daughters, and K_i is the number of kids per individual.

We then would accept hypothesis 1 if β_1 is statistically significantly different from zero and positive,[3] since

$$\text{GEA}(i \mid D_i = 1) - \text{GEA}(i \mid D_i = 0) = \beta_1$$

3 H2: Political Preferences

The second hypothesis to test is:

Having daughters leads to more left-leaning political preferences.

3.1 Variables

We use individuals' own perceptions of their place within the political spectrum by considering the survey's direct question on left-right political preferences, plh0004. The survey question is as follows: *"In politics, people often talk about "left" and "right" when it comes to labeling different political attitudes. When you think of your own political views, where would you classify them?".*

[3] As that would mean that we would reject the alternative hypothesis that $\beta_1 = 0$, i.e. that having at least one daughter has no causal effect on gender equality measured by attitudes.

Answers range from 0 (completely left) to 10 (completely right).

However, the main problem with this variable is that it is only part of the data provided to us in 2005, 2009, 2014 and 2019. For this reason, before running the regression we flag again all negative values of plh0004 as missing since in the sample we have received they represent either [-8] *"Question this year not part of survey"*, [-5] *"Not included in this version of the questionnaire"*, or [-1] *"No answer"*.

3.2 Exact Analysis and Interpretation of Results

We estimate the following regression equation:

$$\text{PP}_i = \beta_0 + \beta_1 D_i + \beta_K K_i + \epsilon_i$$

where is PP_i is the respondent's own perception about her political preferences.

We then would accept hypothesis 2 if β_1 is statistically significantly different from zero, since

$$\text{PP}(i \mid D_i = 1) - \text{PP}(i \mid D_i = 0) = \beta_1$$

and β_1 has a negative sign.

4 H3: Gender Equality as Measured by Outcomes

The third hypothesis to test is:

Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes).

4.1 Variables

The gender literature has not yet reached an understanding of what gender-equitable "outcomes" or "behaviors" are. However, we believe that one of the most important outcomes of gender equality would be a convergence in the amount of care work done by each gender. We follow the definition set out by the UN in its 2024 report "Transforming Care Systems", which states that *"care can be understood as a species activity that includes everything we do to maintain, continue and repair our world, while the four phases of care can be understood as "caring about, taking care of, caregiving and care receiving""* (UN, 2024; p.6). In terms of tasks, care work includes direct care of people (e.g., physical, emotional...) and indirect care (i.e., household chores) (UN, 2024).

The SOEP contains several groups of variables in which individuals report the number of hours spent on housework (pli0043_h, pli0012_h, and pli0016_h), running errands (pli0040, pli0054 and

pli0011), caring for children (pli0044_h, pli0019_h, and pli0022_h), and caring for others in need (pli0046, pli0055, and pli0057). In our opinion, these variables are the closest to the definition of care established by the UN.

To create the total number of hours an individual has spend on care work each week, we:

1. Multiply variables describing the hours spent caring for people during a normal working day (pli0043_h, pli0040, pli0044_h and pli0046) by five, e.g.:

$$\text{pli0043_h} = \text{plh0308_v1} \cdot 5$$

2. Add up the obtained values and add the number of hours of care work on Saturday and Sunday (pli0012_h, pli0016_h, pli0054, pli0011, pli0019_h, pli0022_h, pli0055 and pli0057)

We then differentiate between female (CW_h^F) and male hours (CW_h^M) spend on care work, based on the variable sex. We believe that gender equality would be achieved if both genders would contribute on average the same amount of care work. Hence, our measure of gender equality (based on outcomes) is the absolute difference between male and female care work hours:

$$\text{GEO}_h = |\text{CW}_h^F - \text{CW}_h^M|$$

4.2 Exact Analysis and Interpretation of Results

We estimate the following regression equation:

$$\text{GEO}_h = \beta_0 + \beta_1 D_h + \beta_K K_h + \epsilon_h$$

We then would accept hypothesis 3 if β_1 is statistically significantly different from zero and negative, since

$$\text{GEO}(i \mid D_i = 1) - \text{GEO}(i \mid D_i = 0) = \beta_1$$

5 H4: Gender-Heterogeneous Effect on Gender Equality as Measured by Attitudes

The fourth hypothesis to test is:

Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes).

5.1 Variables

The variables here are the same as for H1, with the addition of gender, which is again measured by the variable sex. We transform sex into the variable M_i to make the interpretation easier:

$$M_i = \begin{cases} 1 & \text{if individual } i \text{ is a man, i.e. sex =1} \\ 0 & \text{otherwise} \end{cases}$$

5.2 Exact Analysis and Interpretation of Results

We estimate the following regression equation:

$$\text{GEA}_i = \beta_0 + \beta_1 D_i + \beta_2 M_i D_i + \beta_3 M_i + \beta_K K_i + \epsilon_i$$

The coefficient of interest here is β_2. The logic is as follows: The effect of having daughters for men is given by:

$$\begin{aligned} \text{GEA}(i \mid D_i = 1, M_i = 1) - \text{GEA}(i \mid D_i = 0, M_i = 1) &= (\beta_0 + \beta_1 + \beta_2 + \beta_3) - (\beta_0 + \beta_3) \\ &= \beta_1 + \beta_2 \end{aligned}$$

The effect of having daughters for females is given by:

$$\begin{aligned} \text{GEA}(i \mid D_i = 1, M_i = 0) - \text{GEA}(i \mid D_i = 0, M_i = 0) &= (\beta_0 + \beta_1) - (\beta_0) \\ &= \beta_1 \end{aligned}$$

Therefore, if the effect for men is "larger", it must hold that

$$\beta_1 + \beta_2 > \beta_1$$

which is equivalent to

$$\beta_2 > 0$$

So we would accept hypothesis 4 if β_2 is statistically significantly different from zero and positive.

6 References

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